Mean Nuclear Radius in Benign vs. Malignant Breast Masses: A Statistical Analysis

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1. Overview

This report analyzes the Breast Cancer Wisconsin (Diagnostic) dataset from the UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic) [1]. The research question was whether mean nuclear radius differs between benign and malignant breast masses and how large that difference is. I used descriptive statistics, visualizations, and Welch's two-sample t-test to answer it.

2. Dataset Description

The dataset contains 569 rows and 32 columns. Each row represents a sample from a breast mass, with features computed from a digitized image of a fine-needle aspirate [1]. The columns include a sample ID, a diagnosis label, and 30 numerical features that summarize 10 characteristics of the cell nuclei. The grouping variable was diagnosis, with 357 benign and 212 malignant samples. The main numerical variable was radius_mean, the mean nuclear radius recorded for each sample.

3. Methods

The source file had no header row, so I constructed the 32 column names from wdbc.names. Before testing, I inspected the data types and checked for missing values, duplicate rows, and duplicate IDs. This follows the broader principle of screening data before formal analysis described by Lusa and colleagues [2]. I used counts and percentages to describe the group sizes, means and medians to compare typical values, standard deviations and quartiles to compare spread, and skewness to assess asymmetry.

The density histogram shows the shape and overlap of the two radius distributions while accounting for unequal group sizes. The boxplot with individual points compares medians, spread, and overlap on the same scale. Following Weissgerber and colleagues [5], I used the histogram and boxplot with points to show distributions of continuous measurements. The Spearman heatmap shows which features rank samples similarly, and I reserved the bar chart for sample counts.

I used the two-sided Welch's t-test as the default for comparing two independent means at the 0.05 significance level because it allows unequal variances [3]. The malignant group's sample variance was 3.24 times the benign group's, illustrating why that flexibility mattered here. The test assumes independent observations, which unique sample IDs alone cannot verify. The large group sizes support an approximately normal sampling distribution of the means despite skewness. I also calculated a 95% confidence interval for the malignant-minus-benign mean difference.

4. Results

The data checks found no missing values, duplicate rows, or duplicate IDs. All ten mean features were right-skewed (0.46 to 1.65), with mean area showing the greatest skew, and their numerical scales varied substantially. Malignant samples had a higher average mean nuclear radius than benign samples (17.46 versus 12.15) and greater spread (standard deviations of 3.20 versus 1.78). The notebook histogram (*Distribution of Mean Nuclear Radius by Diagnosis*) shows that the malignant distribution is shifted toward higher values, with overlap between groups. The boxplot (*Mean Nuclear Radius: Benign*

vs Malignant) best supports the research question because it shows the higher malignant median (17.33 versus 12.20), wider spread, and overlap between individual measurements.

The notebook heatmap (*Spearman Correlation Between the Ten Mean Features*) shows that the correlations among mean radius, perimeter, and area all round to 1.00, meaning those features ranked samples almost identically. The bar chart (*Number of Samples by Diagnosis*) shows that 62.7% of the samples were benign and 37.3% were malignant.

Welch's test yielded $t = 22.21$ and $p < 0.001$, so I rejected the null hypothesis of equal population means at the 0.05 level. The p-value summarizes how incompatible the result is with equal population means under the test assumptions [4]. To report the size and precision of the difference, I estimated the mean gap at 5.32, with a 95% confidence interval of 4.85 to 5.79.

5. Interpretation for a Non-Technical Audience

The average nuclear radius was about 44% larger in malignant samples. If the two groups truly had equal average radii and the test assumptions held, a difference this large relative to its uncertainty would be extremely unlikely by chance. Some benign and malignant samples had similar readings, so a single radius measurement cannot separate every sample in the two groups.

6. Limitations and Potential Bias

Limitation. The analysis compares one feature between groups and does not evaluate diagnostic performance on independent data. A statistically significant difference in group averages does not establish how accurately that feature classifies individual samples.

Potential selection bias. The dataset comes from breast masses sampled by fine-needle aspiration. If the cases selected for that procedure differ systematically from cases seen in other settings, the observed difference in mean radius may not generalize to those settings. The proportion of malignant samples describes this dataset and should not be interpreted as a population prevalence estimate.

The dataset documentation does not specify a physical unit for radius, so the absolute difference is reported on the dataset's measurement scale.

Next, I would examine shape features and test whether a model combining several measurements improves classification. I would validate that model on data from another institution to assess how well it generalizes.

7. References

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